

WINDOWS COMPUTER CLEAN-UP AND VIRTUAL MEMORY PRACTICAL

Department: ICT Department

Practical Title: Windows Update, Cache Cleaning, Temporary File Removal and Virtual Memory Configuration

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Department/Office: _____

Date: 24/7/26

Supervisor: _____

1. PRACTICAL OBJECTIVES

By the end of this practical exercise, the intern should be able to:

- Check the computer's available disk space.
- Create a system restore point.
- remove temporary files.
- Clear Windows cache files.
- Empty the Recycle Bin.
- Clear the DNS cache.
- Run Windows Disk Cleanup.
- Check and repair Windows system files.
- Scan for Windows updates.
- Configure Windows virtual memory.
- Restart and test the computer.
- Record any issues encountered during the exercise.

2. IMPORTANT SAFETY INSTRUCTIONS

- ☒ Confirm that the computer has been approved for maintenance by the ICT Department.
- ☒ Save and close all open documents before starting.
- ☒ Confirm that the user has backed up important files.
- ☐ Do not delete files from the user's Documents, Desktop, Downloads or other personal folders unless instructed by the ICT supervisor.

Completion

☐ Computer information recorded successfully.

Issues encountered

None

PART B: CHECK AVAILABLE STORAGE

Task 2: Check the computer's disk space

Method 1: Using File Explorer

1. Press:
Windows key + E
2. Select **This PC**.
3. Check the free space under **Local Disk (C:)**.

Record the result

Total disk size: 500 GB

Free space before cleanup: 273 GB

Used space before cleanup: ~~27~~ 202 GB

Completion

☒ Available disk space checked and recorded.

Issues encountered

None

Task 3: Check disk space using PowerShell

Step 1

Right-click the **Start** button.

Windows key + R

Step 2

Type:

SystemPropertiesProtection

Step 3

Press **Enter**.

Step 4

Select the **C:** drive.

Step 5

Confirm that protection is set to **On**.

Step 6

Click **Create**.

Step 7

Enter the restore point name:

Before ICT Cleanup

Step 8

Click **Create** and wait for completion.

Completion

☒ System Protection was enabled.

☒ Restore point was created successfully.

☒ Restore point name was recorded.

Issues encountered

None

☒ Files currently in use were skipped.

Issues encountered

None

Task 6: Clear user temporary files using PowerShell

Open PowerShell as Administrator and run:

```
Get-ChildItem -Path $env:TEMP -Force -ErrorAction SilentlyContinue | Remove-Item -Recurse -Force -ErrorAction SilentlyContinue
```

Completion

☒ PowerShell command executed.

☒ User temporary files were cleared.

Issues encountered

None

PART E: DELETE WINDOWS TEMPORARY FILES

Task 7: Clear the Windows temporary folder

Step 1

Press:

Windows key + R

Step 2

Type:

C:\Windows\Temp

Step 3

Press **Enter**.

PART F: EMPTY THE RECYCLE BIN

Task 9: Empty the Recycle Bin manually

Step 1

Locate the **Recycle Bin** on the Desktop.

Step 2

Right-click **Recycle Bin**.

Step 3

Select:

Empty Recycle Bin

Step 4

Click **Yes**.

Completion

☒ Recycle Bin emptied successfully.

Issues encountered

None

Task 10: Empty the Recycle Bin using PowerShell

Run:

Clear-RecycleBin -Force

Completion

☒ PowerShell command executed.

☒ Recycle Bin cleared.

Windows key + R

Step 2

Type:

wsreset.exe

Step 3

Press **Enter**.

Step 4

Wait for the command window to close automatically.

The Microsoft Store may open after the process is completed.

Completion

☒ Microsoft Store cache reset command executed.

☒ Process completed without an error.

Issues encountered

None

PART I: RUN WINDOWS DISK CLEANUP

Task 13: Open Disk Cleanup

Step 1

Press:

Windows key + R

Step 2

Type:

cleanmgr

Step 3

Press **Enter**.

Completion

- ☒ Disk Cleanup opened.
- ☒ Appropriate cleanup categories selected.
- ☒ Downloads folder was not deleted without approval.
- ☒ Disk Cleanup completed successfully.

Issues encountered

None

PART J: CLEAR WINDOWS UPDATE DOWNLOAD CACHE

Task 14: Stop Windows Update services

Open PowerShell as Administrator.

Run:

```
Stop-Service -Name wuauserv -Force
```

Run:

```
Stop-Service -Name bits -Force
```

Completion

- ☐ Windows Update service stopped.
- ☒ Background Intelligent Transfer Service stopped.

Issues encountered

Issue stopping windows update service using code.
"Cannot find any service with service name wuaserv"

Task 15: Delete downloaded Windows Update cache files

Run:

Record the result

- ☒ Operation completed successfully.
- ☒ Corruption was detected and repaired.
- ☒ The command completed with errors.

DISM result or error message:

Result: The restore operation completed successfully

Task 18: Run Windows System File Checker

Run:

sfc /scannow

Wait until the scan reaches 100%.

Record the result

- ☐ Windows Resource Protection did not find integrity violations.
- ☒ Corrupt files were found and repaired.
- ☐ Corrupt files were found but some could not be repaired.
- ☐ The scan did not complete.

SFC result or error message:

Result: Resource protection found corrupt files and successfully repaired them.

PART L: CHECK WINDOWS UPDATES

Task 19: Open Windows Update

Windows 11

Press:

Windows key + I

Issues encountered

None

PART N: RESTART THE COMPUTER

Task 23: Restart Windows

Before restarting:

- ☒ All documents saved.
- ☒ All programs closed.
- ☒ User informed about the restart.
- ☒ Windows Update completed or paused at a safe stage.

Restart using:

`shutdown /r /t 0`

Alternatively:

☒ Start → Power → Restart

Completion

- ☒ Computer restarted successfully.
- ☒ Windows started normally after restart.
- ☒ User successfully logged in.

Issues encountered

None

Task 26: Test basic computer functions

After restarting, test the following:

- ☒ Computer starts normally.
- ☒ Keyboard works.
- ☒ Mouse works.
- ☒ Internet connection works.
- ☒ Office applications open.
- ☒ Printer connection works.
- ☒ Shared folders or network drives work.
- ☒ Antivirus is active.
- ☒ Windows Firewall is active.
- ☒ No unexpected error messages appear.

Issues encountered

None

PART P: FINAL PRACTICAL CHECKLIST

Mark each completed activity.

- ☒ Computer information recorded.
 - ☒ Initial free disk space recorded.
 - ☒ Restore point created.
 - ☒ User temporary files deleted.
 - ☒ Windows temporary files deleted.
 - ☒ Recycle Bin emptied.
 - ☒ DNS cache cleared.
 - ☒ Microsoft Store cache cleared.
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PART R: INTERN'S SUMMARY

1. What activities did you complete?

- | | |
|------------------------|--------------------------------|
| i) Updating the system | iv) Creation of restore points |
| ii) Disk cleanup | v) Backup of data |
| iii) DNS flushing | vi) System scan for errors |

2. How much disk space was recovered?

4 GB

3. What did you learn about virtual memory?

Virtual memory is the memory allocated by the CPU when the RAM is limited

4. Which command was most useful?

sfc /scannow - Windows system file checker

5. What problems remain unresolved?

Stopping and restarting windows update services using powershell command.

PART S: SUPERVISOR'S ASSESSMENT

Assessment Area	Maximum Marks	Marks Awarded
Observed safety and backup requirements	10	
Recorded computer information correctly	10	
Cleared temporary files and caches	20	
Used commands correctly	15	
Ran Windows repair tools	15	